# Working with Data in Excel

*(Extract, evaluate, and present data using formulas and formatting)*

## *Overview*

This course uses an artifact sequence approach in which participants transform a raw employee records dataset into a clean, classified, and summarized workbook. Learning progresses from extracting and cleaning text fields, to calculating date-derived values, to classifying records using logical functions with conditional formatting applied as the visual layer, and finally aggregating results on a formula-driven summary sheet. Each topic builds on the same workbook artifact, enabling participants to reproduce a complete data processing workflow on their own datasets from the following week.

## *Objectives*

Upon completion of the course, participants will be able to process and analyze structured data in Excel by applying text, date, logical, and aggregation functions in sequence. This outcome emphasizes clean inputs, formula-driven classification, and summarized outputs for professional reporting.

Specifically, participants will be able to:

- clean and extract text values using TRIM, LEN, LEFT, RIGHT, and MID
- calculate tenure and age using TODAY, YEAR, MONTH, and DATEDIF
- classify and flag records using IF, AND, and OR, with conditional formatting as the visual layer
- summarize data by condition using COUNTIF and SUMIF on a dedicated summary sheet

## *Who Should Participate*

- Staff who work with employee, customer, or operational data in Excel.
- Analysts who need to clean, classify, and summarize data without pivot tables.
- Professionals building formula-driven workflows from raw data.

## *Key Topics*

I. Cleaning and Extracting Text
   a. Using TRIM to remove extra spaces
   b. Using LEN to verify field lengths
   c. Using LEFT and RIGHT to extract substrings
   d. Using MID to extract from a defined position

II. Working with Dates and Time
   a. Using TODAY to reference the current date
   b. Using NOW to capture date and time
   c. Extracting YEAR and MONTH from a date column
   d. Calculating tenure and age using DATEDIF

III. Evaluating Conditions
   a. Using IF to classify records into categories
   b. Using AND to flag multi-condition records
   c. Using OR to flag any-condition records
   d. Applying conditional formatting visually

IV. Aggregating by Condition
   a. Using COUNTIF to count by a single criterion
   b. Using SUMIF to total by a single criterion
   c. Building a formula-driven summary sheet
   d. Verifying totals against raw data